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A strong, flame-retardant, and thermally insulating wood laminate

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HIGHLIGHTS

- A new strategy to simultaneously enhance the mechanical properties and flame retardance of wood.
- A 3D protective impervious coating as well as a densely packed laminate structure.
- The wood laminate extends the applications of wood into high-performance structural materials.

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ABSTRACT

Developing a strong, flame resistant, and thermally insulating structural material is essential for reducing energy consumption and improving safety in buildings. Wood is a pervasive structural material with natural thermally insulating properties. However, its flammability and insufficient mechanical strength compromises its safety, particularly in mid-rise and high-rise buildings that have higher safety and structural requirements. Herein, we demonstrate a strong, flame-retardant, and thermally insulating wood laminate featuring a compact structure that we synthesized by infiltrating delignified basswood with bentonite nanosheets, followed by hot-pressing to densify the material. Bentonite nanosheets can be infiltrated into the microchannel of the delignified wood to achieve a good compatible and 3D flame resistant protective surface. Flame retardant tests show that the peak heat release rate of this wood laminate is reduced by more than 50% compared to natural basswood. Additionally, the wood laminate does not start to burn until after 7 min of direct exposure to an open flame, and then self-extinguishing when removed from the igniting flame, suggesting the excellent flame resistance of the composite. The wood laminate also exhibits a high tensile strength of 330 MPa that is more than 8-times higher than the starting material. In addition, the wood laminate exhibits lower thermal conductivities of 0.2 and 0.18 $W\,m^{-1}\,K^{-1}$ in the axial and radial direction, respectively. The demonstrated wood laminate shows great promise as a high-performance structural material with strong flame-retardant, thermally insulating, and mechanical robust properties that are needed for the construction of energy efficient buildings.

1. Introduction

The energy consumed in building accounts for more than 30% of the world's total energy consumption, emphasizing the need for more energy-efficient building designs and materials [1–3]. Decreasing heat transfer through the thermal insulation materials can help reduce heating and cooling energy use and may be one of the most effective measures for improving the energy efficiency in buildings [4]. Commercial thermal insulators in the buildings are including plastic foams, extruded polystyrene, and other materials are made from petroleum resources, and can be harmful to health. Furthermore, they are easily ignitable and can sustain a fire due to the lack of fire barriers [5]. In contrast to petroleum-based thermal insulators, wood is a renewable, sustainable, and naturally thermally insulating material that has been used in construction for thousands of years [6–8]. Though wood offers better insulation properties than metal and concrete, currently wood is not as widely used in construction due to its flammability and insufficient mechanical strength [9,10]. As a result, simultaneously improving the flame resistance and mechanical strength of wood while also maintaining its thermally insulative behavior is desirable for energy efficient building applications.

The flammability of wood can be diminished by impregnating or

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coating the material with flame retardants [11]. However, many common flame retardants, such as halogens (brominated flame retardants) and ammonium phosphate, are disadvantages to human health [12–15]. As an alternative, nonhalogenated organic-inorganic nanostructured composites has proven to be an effective way of improving flame retardance [16–21]. The compatibility between inorganic fillers and polymer matrices has a great influence on the flame-retardant properties of materials, in which poor compatibility will lead to catastrophic disintegration and fail to prevent the thermally induced collapse of the building [1]. Wood features a hierarchical structure that presents an abundance of channels (vessels and tracheid lumens) that run along the growth direction [22], which can transport water and minerals [23]. Wood cell walls consist of a variety of components, including cellulose, hemicellulose, lignin [24]. Through chemical modification and physical treatments, this natural structure of wood can be used to fabricate a range of nanostructured multifunctional materials, such as energy storage, structural materials, water treatment, and other applications [25–29].

Herein, we demonstrate a wood laminate featuring excellent flame resistance, thermal insulation, and mechanical strength, made by impregnating bentonite nanosheets into the wood channels. The main component of bentonite is montmorillonite, which is composed of a tetrahedral geometry of SiO_4 on the surface of aluminosilicates, features reduced flammability, heat resistance, and mechanical properties [30]. The structure of bentonite nanosheets is also beneficial for forming hydrogen bonding and enhancing the interfacial interactions with polymer matrices due to the H-bonding networks [31,32]. The thickness of montmorillonite is approximately 1 nm, thus it is feasible to intercalate bentonite nanosheets into wood lumen or nanopores [33]. Additionally, the excellent compatibility between the wood cellulose and inorganic filler helps improve the flame-retardant property of the wood, and the densified wood laminate structure provides enhanced mechanical strength. The inexpensive, nontoxic, and strong wood laminate could meet the requirements for a high strength, flame resistance, terminal insulation materials in energy efficient buildings.

In this work, to optimize the wood structure for desirable properties, we made three modifications to the starting natural basswood material (Fig. 1a). First, we chemically removed the lignin and hemicellulose from the wood structure, isolating the cellulose nanofibers and generating a more porous structure. The THR of cellulose, hemicellulose, and lignin is about 8.6, 4.5, and 4.5 kJ/g [34], respectively. In addition, removing partial compounds from the wood make contribution to the infiltration of bentonite nanosheets in the wood structure. Both the removal of partial components and the infiltration of bentonite may lead to a reduced heat released than its starting natural basswood during combustion. The resulting delignified wood is subsequently immersed in a suspension of bentonite nanosheets to enable infiltration throughout the wood pores. Finally, we hot-pressed the bentonite nanosheets infiltrated throughout of delignified wood to achieve a densely packed laminate structure (Fig. 1b). A weak peak at 848 cm^{-1} corresponding to the Al-O linkage is shifted to 844 cm^{-1} due to the formation of the Al-O-C covalent bonds via the hydroxyl groups of cellulose interacting with the Al on the bentonite platelet surface (Fig. S1) [32,35–37]. The good compatibility between the bentonite nanosheets and cellulose is due to the enhanced hydrogen bonding and the formation of Al-O-C bonds via the hydroxyl groups of cellulose interacting with the Al on the bentonite platelet surface (Fig. 1b) [35].

2. Experimental

2.1. Materials and chemicals

Basswood was purchased from Saunders Midwest LLC. $NaClO_2$ (> 98%, Sigma-Aldrich) was used to remove the lignin and hemicellulose. Bentonite was purchased from Sigma-Aldrich. First, a 6 wt% bentonite dispersion was prepared by stirring at 600 rpm for 7 days, followed by ultrasonication for 30 min. The bentonite suspension was then centrifuged at 7000 rpm for 5 min to remove aggregates and microbubbles. Meanwhile, the lignin and hemicellulose of the natural basswood was removed by immersing the basswood ($100 \times 50 \times 12\ mm^3$ and $100 \times 50 \times 20\ mm^3$) into boiling $NaClO_2$ solution with a pH value of 4.6. After delignification, the resulting delignified basswood material was washed with ethanol and deionized water 5 times to remove residual chemicals. The wet delignified wood was immersed in the prepared bentonite nanosheet solution for bulk infiltration under 200 Pa to remove the ethanol solvent. And then, in

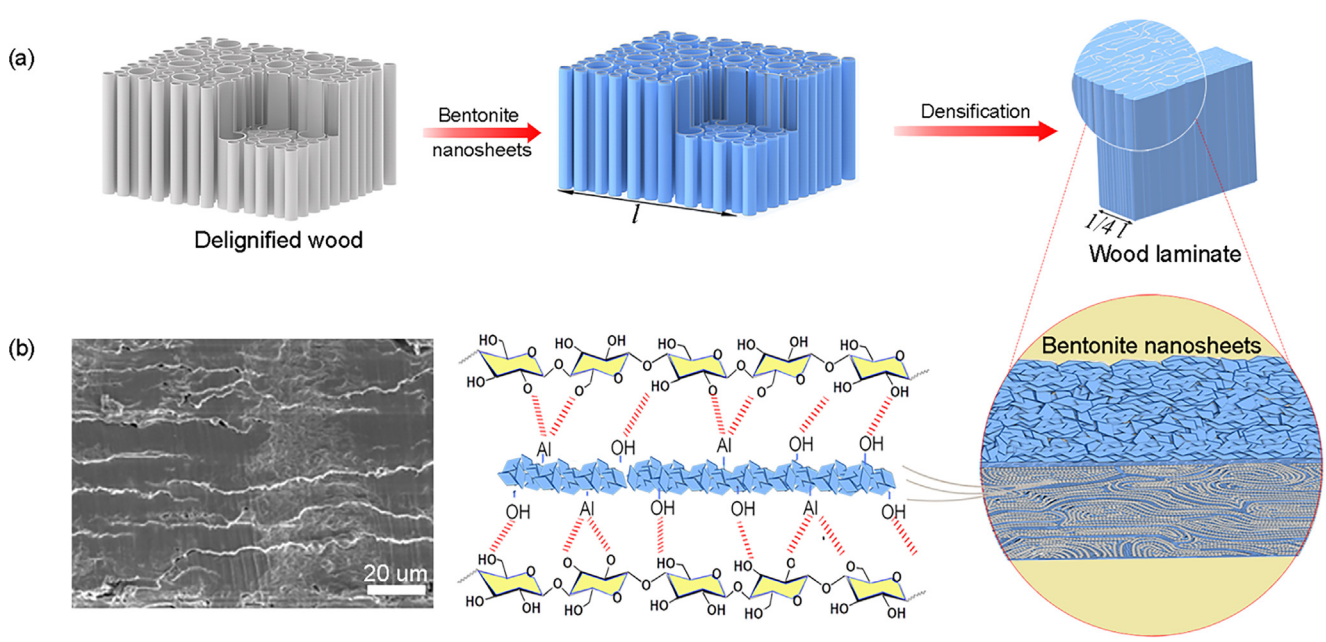


Fig. 1. A flame resistant and thermally insulative wood laminate with bentonite nanosheets. (a) The bentonite nanosheets can be infiltrated into the delignified wood structure due its aligned channels and pores. After hot-pressing, the resulting densified wood laminate consists of cellulose nanofibers and well compatible bentonite nanosheets throughout. (b) The SEM image of the densified wood laminate, and the formation of hydrogen bonding and Al-O-C bonds between the cellulose nanofibers and bentonite nanosheets.

order to ensure the excellent flame resistance, bentonite nanosheets were coated on the surface of the bentonite-infiltrated delignified wood. The final, densified wood laminate was obtained by hot-pressing the wet bentonite-infiltrated delignified wood under a pressure of 5 MPa for 6 days at 100 °C.

2.1.1. Flame-retardant test

The flame retardancy of the wood samples ($100 \times 100 \times 3\,mm^3$) were determined using a cone calorimeter (R-S/FTT0007) under a heat flux of $35\,kW/m^2$ and $50\,kW/m^2$ according to the ISO 5660-1 standard to determine the heat release rate (HRR), effective heat of combustion (EHC), total heat release (THR), and time to ignition (TTI) values of the samples.

2.1.2. Flammability test

The flammability of the natural wood and wood laminate ($100 \times 50 \times 5\,mm^3$) were conducted by a butane torch. The samples were horizontally fixed and ignited by a 4 cm butane flame. The exposure time is up to the flame resistance of samples, we recorded the combustion behavior of the sample until it can burn itself.

2.2. Thermal conductivity measurement

The thermal conductivity along the wood samples was measured using a steady-state method that is similar to that described in our previous research [38–40]. The size of the wood sheet samples was around 20 mm × 35 mm with a thickness of 2 mm. One end of the sample is heated by a laser beam, which acts as the heat source. The other end of the sample is cooled by a heat sink. The sample is placed in a vacuum chamber to minimize the heat loss caused by natural convection. The input power is measured by a laser power meter. The temperature of the sample is measured by a FLIR Merlin MID infrared (IR) camera.

If there is no heat loss caused by natural convection and radiation, the thermal conductivity of the sample can be calculated by Fourier's law:

$$k = \frac{QsL}{A\Delta T}$$

where Q_s is the power flowing through the sample, L is the distance between two points along the direction of the heat flow, A is the measured cross-sectional area, and ΔT is the temperature difference measured along the distance L. However, radiation is inevitable in the measurement. A steady state thermal model, which takes radiation heat loss into account, is built with ANSYS Workbench to obtain the thermal conductivity. The thermal conductivity of the sample is adjusted in the model and the value which achieves the best match between the temperature differences along a distance given by the model and experiment.

2.3. Characterization

The components of the natural basswood and delignified wood were measured according to the National Renewable Energy Laboratory (NREL) method [41]. The morphology and structure of the basswood, delignified wood, and delignified wood with bentonite nanosheets, and wood laminate were observed on a scanning electron microscope (SEM Hitachi SU-70). The size and thickness of the bentonite nanosheets were determined by Atomic force microscopy (AFM, Bruker Dimension Icon). Wood samples were sectioned into samples 7 cm in length and 3 mm in width. The stress-strain measurements of the samples were carried out by a Tinius Olsen H5KT tester. The hardness modulus was measured according to GBT 230.1-2004 with a Rockwell hardness tester (LAISHI Henwall Tech CO. LTD), in which the thickness of the test specimens was 3–4 mm. The thermal stability of the wood samples was determined using thermogravimetric (TG) analysis under nitrogen and air atmosphere from room temperature to 800 °C at a heating rate of $10\,°C{\cdot}min^{-1}$.

3. Results and discussions

3.1. Morphology of the natural wood and flame-retardant wood

For the first step of the synthesis, we used sodium chlorite ($NaClO_2$) to remove lignin and hemicellulose from the natural basswood (see Methods for more details). We note that delignification results in a significant reduction of the lignin. Compared with the natural basswood (20 mm in thickness), lignin content in delignified wood surface and interior area is about 2.3% and 16.7% (Fig. S2), respectively, which is much lower than that of natural basswood (23.6%). After delignification, the natural wood turns white and features a more porous structure (Fig. S3a–f) as a result of the removal of the tight connections between the cellulose nanofibers that compose the wood cell walls [42]. The bentonite nanosheets dimensions were conducted by AFM. The AFM height measurements show a thickness of 1.4 nm for the bentonite particles. The lateral size of bentonite nanosheet was about 40 nm (Fig. S4). In this regard, the increased porosity of the material allows it to be easy infiltrated by a slurry of bentonite nanosheets, which can penetrate throughout the channels of the delignified wood, and engender the resulting composite with its inorganic flame-retardant properties (Fig. 2a). Fig. 2b shows a digital image of the delignified wood after infiltration by the bentonite nanosheets without pressing. Compared with the bentonite-infiltrated delignified wood, SEM images of the natural wood and delignified wood clearly shows a smooth cross section and cell walls, in particular, delignified wood present an increased porosity due to the removal of lignin and hemicellulose (Fig. 2c–f and Fig. S3b, c, e, f). In order to confirm the uniformity of the bentonite nanosheets coating throughout the surface of the wood lumens, we investigated the surface (Fig. 2b A section) and interior (Fig. 2b B section) of the bentonite-infiltrated delignified wood. The high-magnification SEM image in Fig. 2e (surface) shows a rough surface due to the full infiltration of the bentonite nanosheets. In addition, the nanosheets in the cell walls of the bentonite-infiltrated delignified wood are distinguishable, which also indicate that bentonite nanosheets are well attached and infiltrated to the wood cell wall (Fig. 2f). Compared with the cell walls of the delignified wood (Fig. S3e, f), the cell walls of the bentonite-infiltrated delignified wood become thicker and rougher (Fig. S3g–i). We also confirmed that a small amount of bentonite nanosheets attached in the interior section of the wood structure. The phenomenon is attributed to the increased lignin content, resulting in a less porous structure, in which more particles is hard to infiltrate (Fig. 2g, h).

EDX mapping was further conducted to explore the distribution of the bentonite nanosheets. We note that Si, the characteristic element derived from the bentonite nanosheets, is absent in the natural wood starting material (Fig. 2i), while present throughout the bentonite nanosheets-infiltrated wood (Fig. 2j: surface and k: interior). As shown in Fig. 2i–k, EDX mapping visualizes the elemental Si of the cell wall, which suggests that bentonite nanosheets are infiltrated into both the surface and interior wood lumens. After densification treatment of bentonite-infiltrated delignified wood, the open spaces of the wood laminate are disappeared, and the SEM images shows the densely packed cellulose fibers along their growth direction (Fig. S5a), which is different from the channels of natural wood (Fig. S5b). In addition, the surface of the densified wood laminate is covered by a 25-um thick layer of bentonite nanosheets, which should be beneficial for the composite's flame resistance (Fig. S5a).

3.2. Flame retardancy of natural wood and the modified wood laminate

Fig. 3a shows a photograph of the wood laminate, showing a reduction in thickness of 75% compared to natural basswood. We

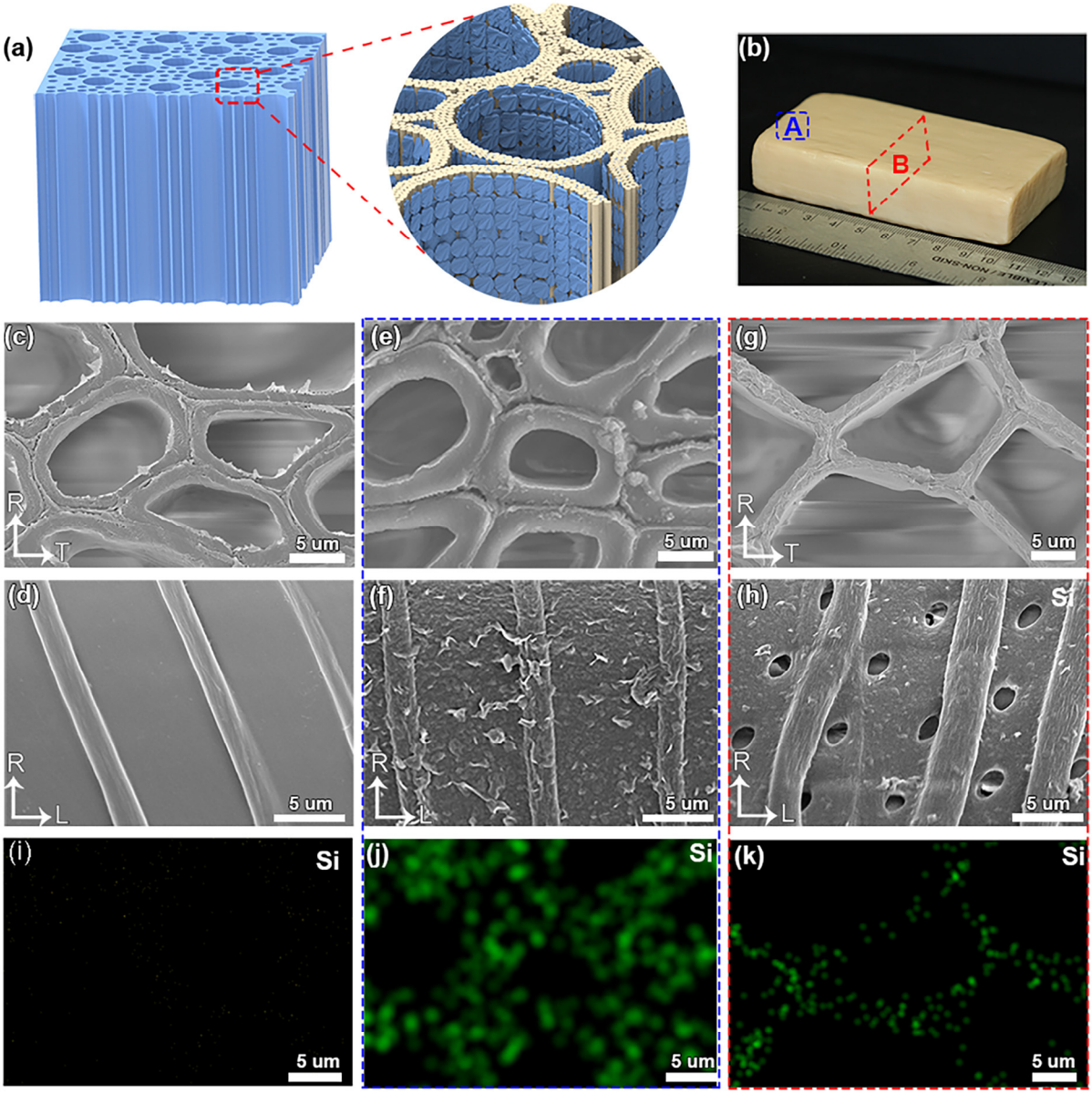


Fig. 2. Schematic illustration the structure and scanning electron microscopy (SEM) images of the bentonite-infiltrated delignified wood and natural wood. a) Schematic of the wood lumens infiltrated with bentonite nanosheets aligned along the wood growth direction. b) Digital image of the delignified wood infiltrated with bentonite nanosheets. c) Cross-sectional high-magnification SEM image of natural basswood, demonstrating its smooth porous surface. d) Side view SEM image of the cell walls, showing clear spirals on the lumen walls. e) Cross-sectional SEM images of the surface region (section A) of the bentonite-infiltrated delignified wood. f) Longitudinal SEM images of the surface region (section A) of the bentonite-infiltrated delignified wood. g) and h) Top view SEM and side view SEM images showing interior section (B) of the bentonite-infiltrated delignified wood. (i) EDX spectroscopy of the natural wood, showing little elemental Si. (j) and (k) EDX spectroscopy of the surface and interior of the bentonite-infiltrated delignified wood, showing the presence of Si due to the infiltrated bentonite nanosheets.

evaluated the flammability of the wood laminate by cone calorimetry under a heat flux of 35 and 50 kW/m^2 in the horizontal position. This measurement provides several important parameters for evaluating fire safety, including the PHRR, TTI, EHC, and THR [43]. We hypothesized that the bentonite nanosheets on the surface and interior of wood laminate would create an impervious carbonaceous char upon exposure to flame, blocking the transport of heat and oxygen into the inner part of the wood structure and thus slowing ignition (Fig. 3b). For comparison, we measured the TTI of the natural wood and the wood laminate at 35 kW/m^2 (Fig. 3c). The TTI of the natural basswood is approximately 35 s, while the wood laminate ignites after 113 s at 35 kW/m^2, respectively. In addition, the EHC of the wood laminate (9.8 kJ/g) decreased by 43% than natural basswood (17.4 kJ/g), indicating the reduction of the heat emission ability of the wood laminate in the process of combustion (Fig. 3c). We attribute the delayed ignition time and reduced heat emission of the wood laminate to its dense structure, combined with the efficient thermal and oxygen barrier of the bentonite nanosheets. Fig. 3d shows that the PHRR value of the wood laminate (175.2 kW/m^2) is much lower than that of natural basswood (351.4 kW/m^2), suggesting that the compact laminated structure and the 3D coating of bentonite nanosheets significantly reduces the PHRR of the wood laminate (50.1% compared to the natural basswood). The HRR curve of the natural wood steeply reaches its maximum value, which indicates that no protective surface char is produced. In comparison, the HRR curve of the wood laminate shows two principal peaks, with the initial peak originating from the oxidation of volatile

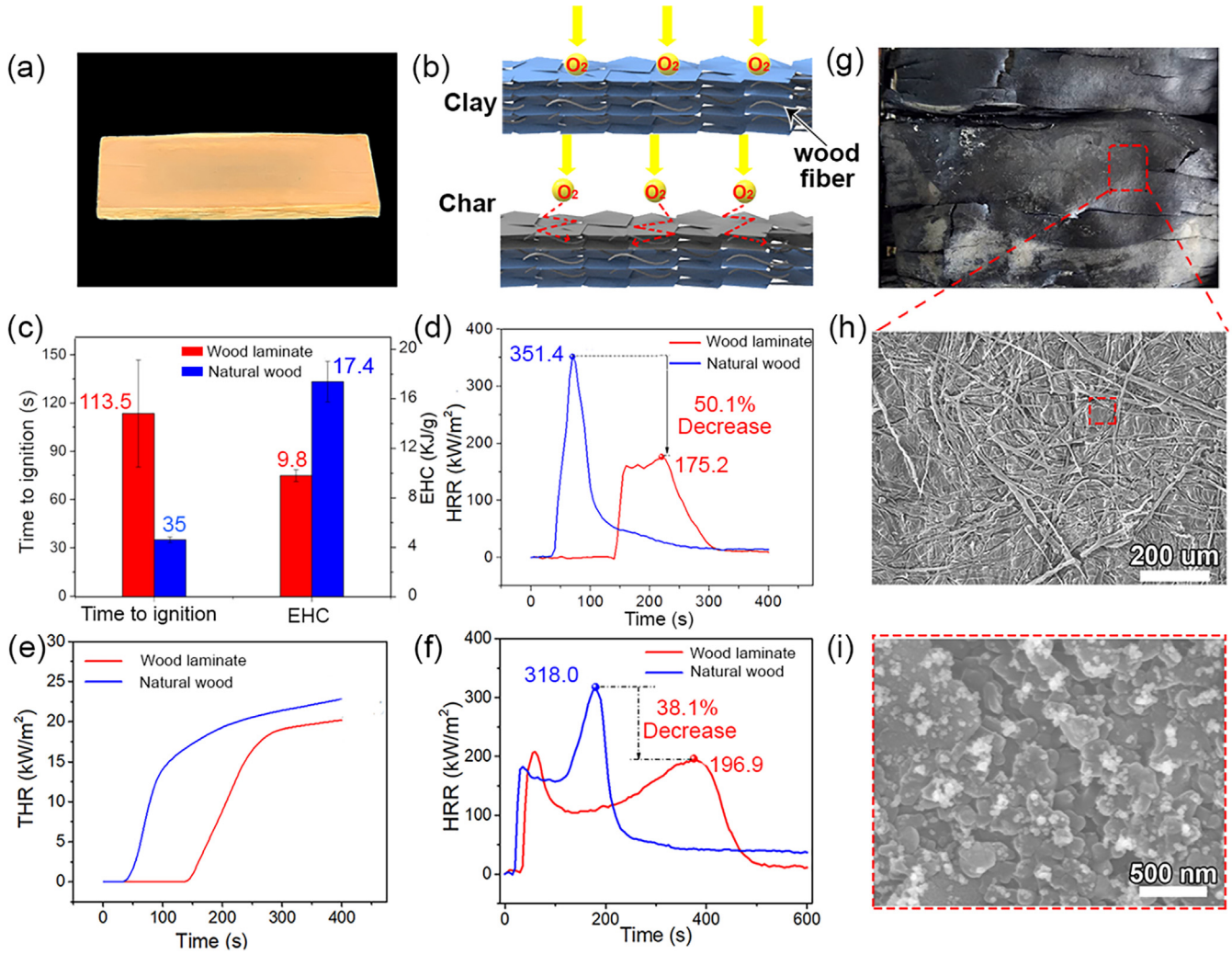


Fig. 3. The flammability of natural wood and the densified wood laminate. (a) Photograph of the prepared wood laminate. (b) Schematic of the combustion behavior of the wood laminate, demonstrating the vital role of the bentonite nanosheets for protecting materials from fire. (c) The TTI and EHC of the natural basswood and wood laminate at a heat flux of 35 kW/m^2. (d) Average HRR and (e) THR curves of the natural basswood and wood laminate. (f) Average HRR curve of the two samples at a heat flux of 35 kW/m^2. (g) Photograph of the wood laminate after combustion, showing the preserved shape of the material. (h) SEM image of the char residue of the wood laminate, showing the carbonized cellulose of wood laminate. (i) A magnified SEM image of the surface of the wood laminate, which consists of small lamellas on the surface cellulose.

pyrolysis products [44]. As burning continues, an insulating surface char is generated on the wood laminate due to the presence of the bentonite nanosheets, and the char acts as a thermal and oxygen barrier to lower the HRR (Fig. 3d, red line). Bentonite nanosheets can act as a thermal insulator and provide catalytic sites, which can enhance degradation paths toward char formation [45,46]. Bentonite nanosheets not only benefit char formation, but also increase the tortuous path for the oxygen diffusive process, which impedes the escape of volatiles and contributes to the flame resistance of the wood laminate [33,47].

THR is another important parameter for evaluating the influence of the fire hazard. A good flame retardant filler should be equipped with the character of reducing the THR of composites effectively [48]. To evaluate the fire behavior of these materials, we studied the same thickness of the natural wood and wood laminate. Note that the original thickness of wood laminate before densification is about 4 times larger than natural basswood, the laminate may show an increase in THR because it contains more cellulose. However, as it can be seen in Fig. 3e, the THR of the wood laminate is similar to the natural basswood. The wood laminate features comparable burning heat potentially due to the removal of lignin and hemicellulose. The excellent flame retardancy of the wood laminate emphasizes how the bentonite nanosheets play a crucial role in promoting char formation and slowing down fire spread. In addition, we subjected the laminate and natural basswood to a constant heat flux of 50 kW/m^2, with the resulting HRR curves of the laminate also showing two principal peaks (Fig. 3f). The first peak is produced due to the charred protective layer. The first peak is due to a charred protective layer of cellulose, which is produced by bentonite nanosheets. The first peak is slightly higher than that of natural wood (a protective char from charred lignin), and the second peak (PHRR) of wood laminate is 38.1% lower than natural wood. The result is attributed to that the charred protective layer caused by bentonite nanosheets is hard to break with a reduced combustion kinetics.

We also note that the TTI of the wood laminate (34 s) is still longer than the natural basswood (12 s) at a heat flux of 50 kW/m^2 (Fig. S6). A photograph of the char residue of the wood laminate after combustion is shown in Fig. 3g. SEM images indicates that a char layer was formed, the wood cellulose was preserved, and the wood laminate can keep a complete retention of the original shape after burning (Fig. 3h). The high-magnification SEM images confirmed the carbonaceous char is reinforced by the small lamellas of bentonite. These lamellas limited oxygen transport to the inner cellulose, creating an excellent physical barrier for the wood laminate and improving the fire-retardant

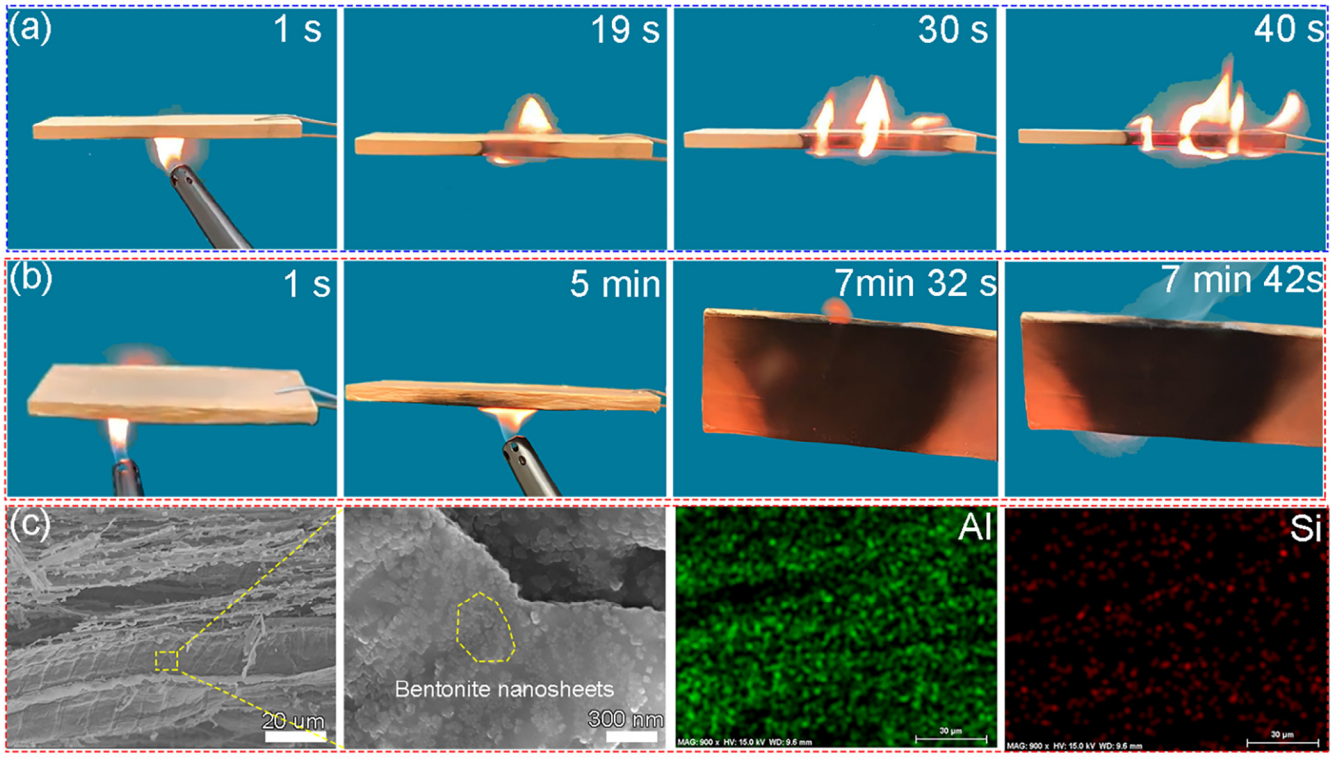
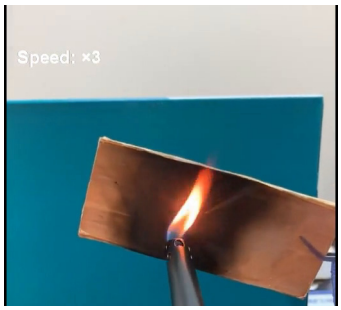


Fig. 4. (a) In the flammability test of the natural basswood, the material begins to burn within 19 s. (b) The wood laminate demonstrates excellent flame resistance, only starting to burn after 7 min. (c) SEM images and EDX mapping of the wood laminate after burning, showing the preserved hierarchical structure due to the presence of the bentonite nanosheets in the infiltrated wood structure.

performance (Fig. 3i).

3.3. Flame retardant test

In order to evaluate the burning behavior of the wood laminate and natural wood, we exposed the materials to an open flame. As shown in Fig. 4a, natural wood ignited within 19 s, resulting in a strong flame at 30 s. In contrast, the wood laminate remained inert and maintained its structure and shape even after being exposed to the torch flame for more than 7 min (Fig. 4b). We attributed the excellent flame resistance of the laminate to its inorganic framework that is composed of a high content of bentonite nanosheets (16.2 wt%). The incorporation of bentonite nanosheets not only leads to a delayed time to ignition, but also contributes to a self-extinguishing effect. As shown in Fig. 4b, the wood laminate began to burn at 7 min and 32 s, but quickly self-extinguished after 10 s when removed from the igniting flame (Video S1). The time to ignition of the wood laminate is the longest of that reported for other fire-resistant wood composites, such as wood/poly (furfuryl alcohol)/ammonium dihydrogen phosphate (TTI 12s) [11], and other wood structural materials [49]. To further investigate the structure evolution after burning, we performed SEM measurements of the inner structure of the burned wood laminate. As is shown in Fig. 4c, the hierarchical wood structure was preserved after burning, and the bentonite nanosheets remained in the cellulose fibers aligned with the wood growth direction. EDX mapping of the wood laminate further verified the presence of Al and Si elements, which are derived from the bentonite nanosheets. The heat shielding properties of the wood laminate benefits from the 3D volumetric infiltration of the bentonite nanosheets into the wood structure. Additionally, the densely packed structure of the wood laminate also contributes to the enhanced flame resistance, because the lack of bulk channels can reduce the concentration of oxygen necessary for fire to burn [49].



Video S1.

3.4. Thermal properties and mechanical properties of wood laminate

In order to make sure the wood laminate is an efficient energy-saving structural material, we further investigated the mechanical properties and thermal management capabilities of the natural basswood, delignified wood, and wood laminate materials in the axial and radial directions. The delignified wood with less lignin and hemicellulose reduces the linkage between the cellulose fibrils [50], resulting in a lower thermal conductivity of the delignified wood ($0.16 \pm 0.02\,W\,m^{-1}\,K^{-1}$ in the axial direction and $0.08 \pm 0.02\,W\,m^{-1}\,K^{-1}$ in the radial direction) (Fig. 5a). Fig. S7a–f shows infrared images of the heat transfer process in the natural wood and delignified wood, in which the elliptical shapes of the temperature gradient profiles indicate the anisotropy of the thermal conductivity in the axial and radial directions of these materials. The measured thermal conductivity of the wood laminate of $0.20 \pm 0.02\,W\,m^{-1}\,K^{-1}$ (in the axial direction) was lower than that of the natural wood $0.21 \pm 0.02\,W\,m^{-1}\,K^{-1}$ (Fig. 5a). In comparison, the thermal conductivity of the wood laminate is $0.18 \pm 0.02\,W\,m^{-1}\,K^{-1}$ in radial direction (Fig. 5a), which is a little higher than that of natural basswood

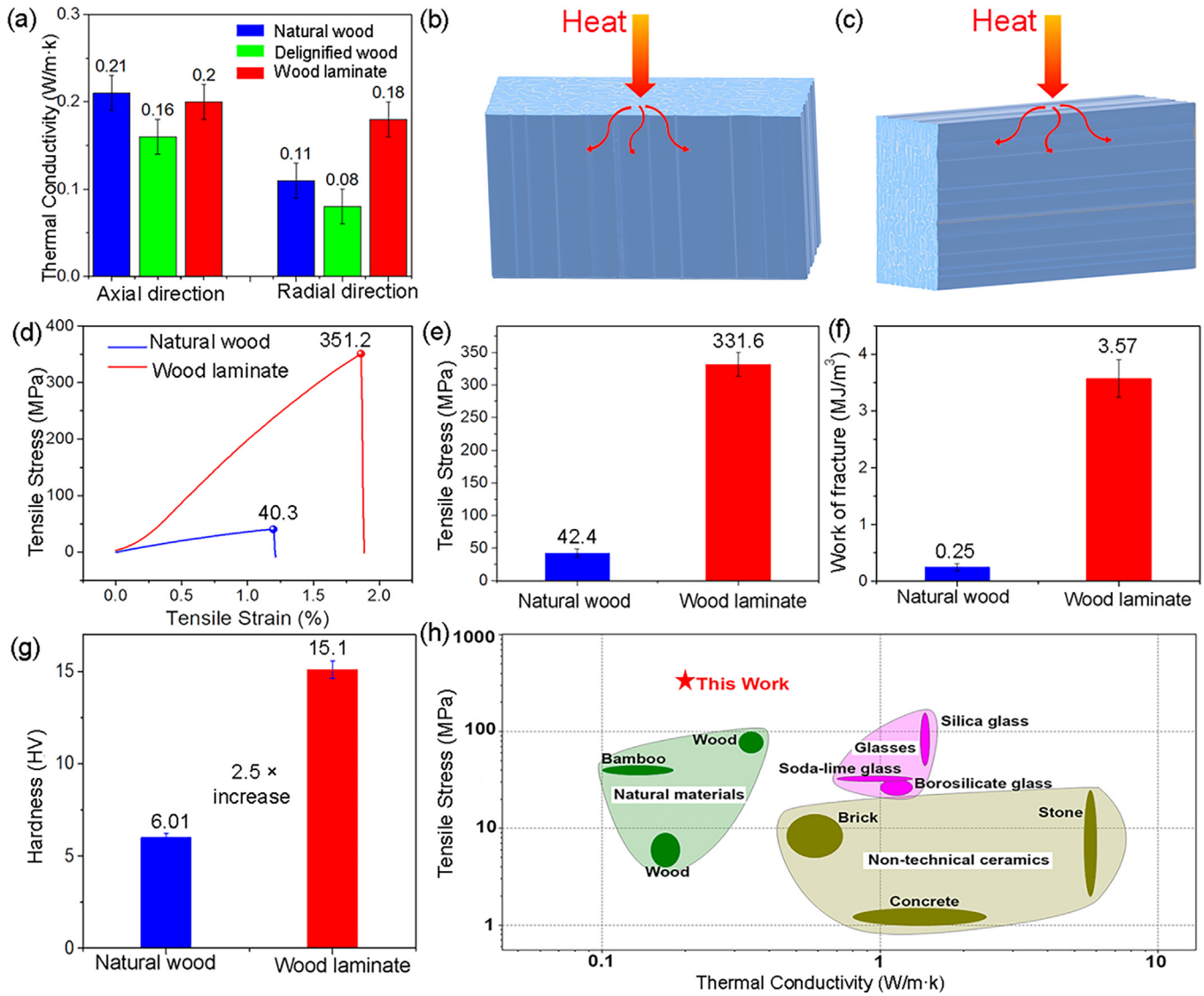


Fig. 5. (a) Comparison of the thermal conductivity of the natural wood, delignified wood, and wood laminate. (b) Schematic diagram of heat conduction in the wood laminate along the axial directions as axial heat transfer and (c) Schematic diagram of heat conduction in the wood laminate radial directions as transverse heat transfer. (d) Tensile stress-strain curves of the basswood and densified wood laminate. (e) Comparison of average tensile strength of natural wood and wood laminate. (f) Comparison of the work of fracture of the natural basswood and wood laminate. (g) The hardness of the natural wood and wood laminate. (h) The wood laminate features both good tensile strength and low thermal conductance compared with other traditional structural building materials.

$0.11 \pm 0.02\,W\,m^{-1}\,K^{-1}$. The schematics (Fig. 5b, c) shows heat transfer process in parallel with the wood laminate cellulose fibers and across the wood laminate growth direction. The thermal conductivity of wood laminate in different directions and the shape of the temperature gradient profile confirm that the anisotropic thermal conduction of the wood laminate is reduced, due to the lower porosity of the densified wood structure after hot-pressing (Fig. S7g, h).

The wood laminate also exhibited superior mechanical performance than that of the natural basswood. The wood laminate features outstanding tensile strength of 331 ± 18.1 MPa, which is 8-times higher than the natural basswood (42.4 ± 6.2 MPa, Fig. 5d, e). The wood laminate (bentonite content 16.2 wt%) demonstrates a density of $0.915\,g\,cm^{-3}$ and a specific strength of ~ $366\,MPa\,cm^{-3}\,g^{-1}$, which is also higher than that of natural wood ($0.448\,g\,cm^{-3}$) with specific strength of $95.3\,MPa\,cm^{-3}\,g^{-1}$ (Fig. S8). As shown in Fig. 5f, the enhanced tensile strength is accompanied with a large increase in the work of fracture of the wood laminate ($3.57 \pm 0.33\,MJ\,m^{-3}$), which is 14-times higher than that of the natural basswood ($0.25 \pm 0.06\,MJ\,m^{-3}$). We attribute the high mechanical strength of the wood laminate to the strong interactions provided by hydrogen bonding and increased Al-O-C bond formation between the wood laminate cellulose and bentonite nanosheets [51]. Compared to the hardness of basswood (6.01 ± 0.25 HV), the wood laminate features an increased hardness of 15.11 ± 0.48 HV, which is due to that the large pores in the wood structure are eliminated after densification (Fig. 5g). Fig. 5h indicates that the wood laminate exhibits an excellent integrated mechanical and thermal insulation properties, with superior tensile strength compared with common structural materials in the construction market, such as concrete, brick, stone, natural wood, and glass.

We further investigated the thermal stability of the natural wood and wood laminate by thermogravimetric analysis (TG) under a nitrogen atmosphere (Fig. S9). As shown in Fig. S9a, the final residual mass of the neat bentonite, delignified wood, and natural basswood is 82.3%, 17.5%, and 3.5%, respectively. However, the residual mass of the wood laminate (41.7 wt% at 800 °C) is significantly higher than that of the delignified wood and basswood. The result is not simply due to the present of bentonite nanosheets (16.2 wt%), in fact, bentonite

nanosheets play an important role to increase the char production, and then enhance the thermal stability [52]. DTG curves (Fig. S9b) show the abrupt weight loss of the natural wood at 355 °C and the major degradation of the wood laminate occurring at 329 °C. The TG curves under air atmosphere show similar results. The residual mass of the wood laminate is about 20.5 wt% at 800 °C (Fig. S9c). In addition, the maximum weight loss temperature of wood laminate is around 320 °C (Fig. S9d), which is higher than that of natural basswood (304 °C). Our TG results both in N_2 and air atmosphere suggest that the incorporation of bentonite nanosheets into the wood structure improves the thermal stability of the wood laminate. We attribute the decreased degradative temperature of the laminate to the delignification of the wood [53]. Our TG results suggest that the incorporation of bentonite nanosheets into the wood structure improves the thermal stability of the wood laminate [54]. Water resistance is a long-standing issue for cellulose-based materials. Cellulose accounts for nearly 40% of wood, the swelling of wood is inevitable without other measures. Our experimental results show that surface coating is a good way to improve the stability against water (demonstrated by the stable weight and dimensions) for the wood laminate (Fig. S10).

4. Conclusion

In summary, we report a strong, fire-retardant, self-extinguishing, and thermally insulating wood laminate comprised of hierarchically aligned bentonite nanosheet-infiltrated cellulose nanofibrils for energy efficient buildings. Delignification significantly increases the porosity of the wood cell walls, enabling the impregnation of bentonite nanosheets throughout the substrate. Subsequent densification creates a 3D protective impervious coating as well as a densely packed laminate structure, which both contribute to the excellent flame-retardant properties of the composite. When exposed to an open flame, the wood laminate does not burn for up to 7 min, which is among the best flame-retardant behavior compared to previously reported fire-retardant wood materials. The mechanical tensile strength of the wood laminate also substantially improved by 8-fold due to the enhanced hydrogen and Al-O-C bonding between the bentonite nanosheets and nanocellulose, and the highly aligned, densely packed cellulose nanofibers. More importantly, the wood laminate presents low thermal conductivities both in the axial and radial directions—superior to other commercial structural materials. These combined merits demonstrate the wood laminate as a strong, flame-retardant, and thermal insulating structural material for energy efficient building applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

G. Chen and C. Chen contributed equally to this work. L. Hu, G. Chen, and C. Chen conceived the idea and designed the experiments. G. Chen contributed to the material fabrication. Y. Pei and B. Yang conducted the thermal conductivity measurement and analyzed the corresponding results. Y. Liu, C. Chen, and G. Chen were responsible for the SEM images. G. Chen, S. He, and C. Chen were responsible for the mechanical tests and FT-IR measurements. C. Chen, G. Chen, S. He, B. Jiang, W. Gan, and D. Liu contributed to the discussion and characterization of the wood laminate. L. Hu, G. Chen, and C. Chen co-wrote the manuscript. All authors commented on the final manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2019.123109.

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